

Technische Universität Chemnitz

Exploring Reinforcement learning strategies in automated wheel truing

Master's Thesis

submitted by

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Abstract

This thesis attempts to find a solution to automatic spoked wheel truing through reinforcement learning in a simulated environment. We test and evaluate a set of different algorithms, state-spaces and reward schemes.

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Chapter 1

Introduction

1.1 Motivation

short motivation preferably relating the problem to material stress optimization problems.

1.2 Wheel truing

describing the problem and mention sim-to-real outlook

1.3 Thesis Structure

This thesis is structured as follows: Chapter 2 reviews related work.

Chapter 2

Theory

This chapter should outline the necessary theoretic background on deep-reinforcement-learning.

2.1 Reinforcement learning basics

Here I want to cover the basic ideas, definitions and mathematical prerequisites. This will probably include:

- Markov states
- states, actions and their values
- value iteration and policies
- function approximation and deep-learning
- data sampling and replay buffer

After these basics, I want to describe the main ideas and various improvements of the different approaches of the three selected algorithms in the next subsections:

2.2 DeepQ (Rainbow)

2.3 Policy gradients (PPO)

2.4 World-Models (TDMPC)

2.5 Wheel Truing Physics

Chapter 3

Environment

Here I want to describe the process and considerations taken to transform the wheel-simulation into a suitable reinforcement learning environment. Loosely, this will include:

- Isolating and optimizing the necessary physics calculations
- State-space design ([n]rimpoints, spoketensions and fourier coefficients)
- action-space formulation (continous onehot, discrete and allspokes)
- reward function design
- episode structure and termination conditions

Chapter 4

Experiments and Methodology

Roughly, a short description of the experimental design would be as follows:

I plan to do two phases of experiments. In phase 1, the goal is to establish a baseline for the different algorithms. I do this by running single training experiments on all the combination of state-, action and reward schemes as well as two network sizes. (maybe without tan displacement and with one state with reduced point numbers)

TODO: justify the parameters used here for each algorithm

- here list the problems (some experiments done with insufficient logging and states i dont want to keep as well as non integrated penalization and maybe bad stopping condition)

TODO: look through the experiments already done and list what to keep and what needs to be redone.

Phase 2 should include a hyper-parameter optimization of the most promising candidates from phase 1.

In this chapter I also want to talk about the evaluation metrics.

TODO: is there something else i want to look to better evaluate?

Chapter 5

Results and Analysis

5.1 Baseline Performance

- performance of the algorithms in phase 1
- what can we say about the influence of reward functions and
- state space
- action space
- cross factor analysis

5.2 hyper-parameter optimization

I am not sure yet which candidates will do well and what exactly I want to evaluate here after optimizing the best candidates.

Chapter 6

Discussion and Conclusion

Appendix

.1 Additional Figures

Additional figures that didn't fit in the main chapters.

.2 Additional Tables

Additional tables with detailed data.

.3 Code Listings

Extended code listings that are too long for the main text.

.4 Experimental Details

Detailed experimental configurations and parameters.